

Performance Metrics for Clustering

1. Introduction

Clustering evaluation differs fundamentally from regression and classification because, in most cases, there are no ground-truth labels. Therefore, clustering metrics aim to measure:

- **Compactness:** How close data points are within a cluster
- **Separation:** How far apart clusters are from each other
- **Agreement:** How well clustering matches true labels (if available)

These metrics are broadly divided into:

- **Internal Metrics:** Do not require ground truth labels
- **External Metrics:** Require ground truth labels

2. Silhouette Score

Definition

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (1)$$

Explanation

- $a(i)$: Average distance of point i to points in the same cluster
- $b(i)$: Minimum average distance of point i to points in other clusters

Interpretation

- $s(i) \approx 1$: Well clustered
- $s(i) \approx 0$: On boundary
- $s(i) < 0$: Misclassified

Insight

Combines cohesion and separation in a single metric.

3. Rand Index (RI)

Definition

$$RI = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

Explanation

Measures similarity between predicted clustering and true labels using pairwise agreements.

Limitation

Does not account for agreement due to chance.

4. Adjusted Rand Index (ARI)

Definition

$$ARI = \frac{RI - \text{Expected RI}}{\text{Max RI} - \text{Expected RI}} \quad (3)$$

Interpretation

- 1: Perfect clustering
- 0: Random clustering
- < 0: Worse than random

Insight

Corrects Rand Index for random chance.

5. Mutual Information (MI)

Definition

$$MI(U, V) = \sum_{i,j} P(i, j) \log \left(\frac{P(i, j)}{P(i)P(j)} \right) \quad (4)$$

Explanation

Measures how much information is shared between true labels and predicted clusters.

Limitation

Not normalized, making comparisons difficult across datasets.

6. Calinski-Harabasz Index

Definition

$$CH = \frac{\text{Between-cluster variance}}{\text{Within-cluster variance}} \quad (5)$$

Explanation

Evaluates clustering based on ratio of separation to compactness.

Interpretation

Higher values indicate better clustering.

7. Davies-Bouldin Index

Definition

$$DB = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{d(c_i, c_j)} \right) \quad (6)$$

Explanation

- σ_i : Intra-cluster distance
- $d(c_i, c_j)$: Distance between cluster centers

Interpretation

Lower values indicate better clustering.

8. Comparison of Metrics

Metric	Type	Measures	Ideal Value
Silhouette Score	Internal	Cohesion + Separation	Close to 1
Rand Index	External	Pairwise Agreement	Close to 1
Adjusted Rand Index	External	Chance-corrected Agreement	Close to 1
Mutual Information	External	Shared Information	High
Calinski-Harabasz	Internal	Variance Ratio	High
Davies-Bouldin	Internal	Cluster Similarity	Low

9. Conclusion

Clustering metrics provide different perspectives:

- **Silhouette**: Individual point quality
- **Rand / ARI**: Pairwise agreement

- **Mutual Information:** Information theory perspective
- **CH Index:** Variance-based separation
- **DB Index:** Cluster overlap

Selecting the right metric depends on whether ground truth labels are available and the structure of the data.